

PRODUCT OVERVIEW · HOW IT WORKS

KnowLiq

An AI Learning Platform Built Around the Student's Learning Journey

"Getting an answer is not the same as learning."

KnowLiq turns AI from an answer provider into a personalized learning system.

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1. The Problem KnowLiq Is Trying to Solve

Traditional studying has a major organizational problem. Imagine a student preparing for an examination. They have lecture slides, textbooks, notes, past papers, assignments, solved examples, and dozens of concepts to remember.

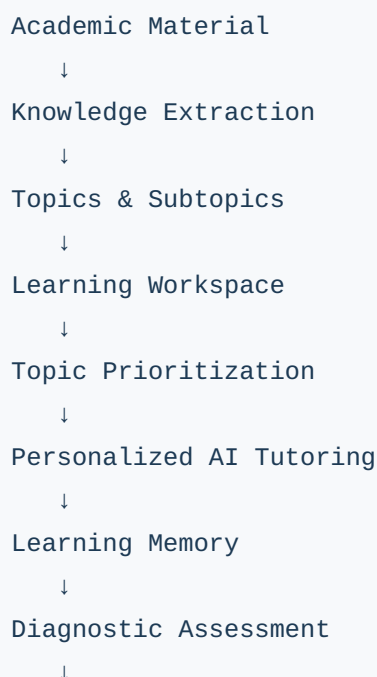
The student has to manually figure out: What is important? What should I study first? What have I already mastered? What have I forgotten? What am I weak at? When should I revise it? Am I actually ready for the examination?

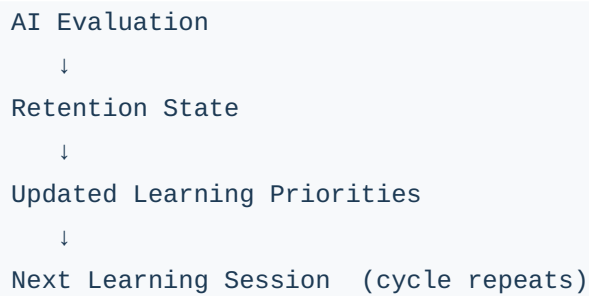
AI can answer individual questions, but answering individual questions does not automatically create a complete learning strategy. For example, a student might ask an AI to explain Bernoulli's equation, then ask for an example, then ask it to solve a problem — but there is no guarantee the student actually understands Bernoulli's equation. The student may leave the conversation thinking "I understand this." Then the examination arrives, and the problem becomes obvious.

Understanding an explanation is not the same as being able to demonstrate understanding. KnowLiq is designed to address this gap.

2. The Core Idea Behind KnowLiq

KnowLiq is built around a continuous learning loop, where the result of one learning session influences the next:





3. Understanding the Student's Academic Material

The process begins when the student provides academic material — lecture notes, course PDFs, past papers, examination papers, academic documents, or other study resources. KnowLiq processes the uploaded material and attempts to identify the academic concepts contained inside it, breaking the document into meaningful topics and subtopics rather than treating it as one large piece of information.

For example, a Fluid Mechanics resource could produce a structure similar to:

- **Fluid Mechanics**
 - **Fluid Properties** — Density, Viscosity, Specific Gravity
 - **Fluid Statics** — Pressure, Hydrostatic Forces, Manometers
 - **Fluid Dynamics** — Continuity Equation, Bernoulli Equation, Momentum Equation

This creates the foundation of the student's learning workspace.

4. Creating a Knowledge Map

One of KnowLiq's core concepts is the **knowledge map**, which represents the academic structure of the material. For each identified topic, KnowLiq can maintain: topic name, subtopics, importance score, number of appearances, associated marks, and learning state.

This turns unstructured academic material into a structured map of things the student needs to learn — far more useful than simply storing the uploaded PDF.

5. Why Past Papers Are Particularly Useful

Past examination papers show which topics appear repeatedly, which concepts receive more marks, which areas are commonly tested, and how the course is actually assessed. KnowLiq extracts this information while processing the material:

Topic	Importance	Appearances	Marks
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Bernoulli Equation	High	5	20
Continuity	High	4	15
Hydrostatics	High	4	18
Fluid Properties	Medium	2	8

This does not mean the system predicts the exact future examination — it gives the learning system useful evidence about the academic importance of different areas.

6. The Learning Workspace

After processing the material, KnowLiq creates a workspace for the subject — the student's persistent academic environment. It can contain the subject, topics, subtopics, importance information, marks, learning state, learning history, retention information, and test information.

The student's academic context is not supposed to disappear when a single chat ends — the workspace is the foundation for future learning sessions.

7. KnowLiq Tracks More Than What You Studied

A core idea in KnowLiq is the distinction between *"The student studied this"* and *"The student understands this"* — these are not the same thing. A student might spend an hour reading about thermodynamics without actually being able to solve a problem. KnowLiq therefore maintains learning states for subtopics, using three main retention states (plus an initial untested state for new topics):

Green — Strong Understanding

The student demonstrated a strong understanding of the topic.

Yellow — Partial Understanding

The student understands the main concept but still has gaps, weak reasoning, or incomplete understanding.

Red — Weak Understanding

The student demonstrated significant misunderstanding, incorrect reasoning, or inability to apply the concept.

8. Topic Prioritization

Once KnowLiq knows the student's topics and learning states, it can determine what deserves attention. Suppose a student has ten subtopics: five strong, three partially understood, two weak. A conventional study system might simply display all ten. KnowLiq instead prioritizes them based on:

1. Red or weak topics
2. Yellow or partially understood topics
3. Green topics that have become due for revision
4. Untested topics
5. Academic importance
6. Marks
7. Time since the topic was last learned

This creates a focused learning session — instead of asking "What do you want to study?", the system can help answer "What would be most useful for you to study next?"

9. Why Topic Prioritization Matters

Students often study inefficiently. They may repeatedly review concepts they already understand because those concepts feel comfortable, while avoiding difficult topics that require more effort. This creates a dangerous situation: the student feels productive without actually improving their weakest areas. KnowLiq's prioritization system reduces this problem by directing attention toward areas that need it most.

10. Personalized AI Tutoring

After determining the current learning focus, KnowLiq uses an AI tutor to teach the student. The tutoring system considers current learning topics, the student's learning preferences, recent concepts, relevant previous conversations, subject, preferred language, learning style, and desired response structure.

The tutor is instructed to avoid overwhelming the student with information. Instead, it can explain one concept at a time, use examples, adapt explanations when the student is confused, use analogies, ask questions, move forward when understanding is demonstrated, remain within the relevant syllabus, maintain the student's selected language, and format mathematical content appropriately.

11–13. Three Different Learner Types

KnowLiq allows students to select how they prefer to learn, and this setting can be changed at any time.

Visual & Analogy Mode

Designed for students who understand abstract concepts better when connected to physical or real-world systems. For example, an engineering concept could be explained through engines, plumbing systems, mechanical systems, or everyday physical processes. Instead of only giving a formal definition, the AI tries to build an intuitive mental model:

Abstract concept → familiar real-world system → intuitive understanding

Interactive Socratic Mode

Instead of making the student a passive receiver of information, the tutor guides them through questions. For example, instead of immediately saying "The pressure increases because...", the AI might ask "What do you think happens to the weight of the liquid above a point as you move deeper?" The student thinks, the AI responds, the student reasons again — an active feedback loop encouraging reasoning, recall, active participation, critical thinking, and conceptual understanding rather than passive reading.

To-The-Point Mode

Designed for students who prefer high information density and minimal unnecessary explanation. Responses focus on key concepts, important facts, bullet points, direct explanations, and minimal fluff. The implementation targets keeping responses under approximately 150 words where appropriate — useful for revision, quick concept checks, familiar topics, last-minute preparation, and students who prefer concise explanations.

14. Field of Study

KnowLiq distinguishes between two broad fields of study:

Technical & STEM

Physics, Chemistry, Mathematics, Thermodynamics, Engineering, and other technical subjects. These often involve formulas, calculations, numerical problems, and technical reasoning.

Theory & Humanities

Biology, Psychology, Philosophy, History, and other theory-oriented subjects. These often require stronger conceptual understanding, explanation, comparison, interpretation, and recall.

This classification gives the AI additional context when generating explanations and assessments.

15. Response Structure Settings

Students also control how detailed the AI's responses should be, with two current options:

Detailed

Provides a more comprehensive explanation — useful when learning a difficult topic from the beginning. For example, "Teach me thermodynamic entropy from the basics" can yield definitions, intuition, relationships, examples, mathematical explanation, and applications.

Normal

Keeps responses more compact while maintaining answer quality, and is designed to reduce AI usage cost by approximately 30%. Useful when a huge response would be unnecessary — e.g. "What is the unit of pressure?"

16. Language Support

KnowLiq supports multilingual AI learning. The current language configuration includes:

English, Spanish, French, German, Italian, Portuguese, Dutch, Russian, Chinese, Japanese, Korean, Arabic, Hindi, and Urdu.

The selected language becomes part of the tutor's response configuration, letting the student learn in their preferred supported language rather than being forced to study everything in English — particularly useful for students who understand difficult academic concepts more naturally in their native language.

17–19. AI Model Architecture

Stella — GPT-5.6 Luna

KnowLiq's higher-capability AI tutor, intended for situations where stronger reasoning and deeper explanations are useful: difficult mathematics, engineering problems, complex scientific concepts, advanced reasoning, and challenging technical questions. The goal is to use a more capable model when the learning task requires it.

Astra — GPT-4o-mini

KnowLiq's lightweight AI tutor, suitable for routine academic interactions such as straightforward explanations, simple concept questions, revision, general tutoring, and routine learning conversations. Using a smaller model for simpler interactions makes the platform more resource-efficient.

Auto Mode

Currently defaults to GPT-4o-mini to conserve resources. However, the architecture can evolve so that Auto analyzes the student's task and selects a more appropriate model automatically:

- Simple concept → Lightweight model
- Complex mathematics → Strong reasoning model
- Programming → Coding-focused model
- Engineering → Model better suited to technical reasoning
- General theory → Efficient general model

This could create a model-routing system where the student does not need to manually decide which AI model should handle every question.

20–23. Learning Memory & Current Focus

A student's learning journey does not exist in one conversation — they may study a topic today and return to it days later. KnowLiq maintains learning information across conversations through several components:

Recent Conversation Context

Recent chats help the tutor understand what has just been discussed.

Semantic Memory

Older conversations can be searched based on their meaning. KnowLiq uses OpenAI's `text-embedding-3-small` model to create embeddings for conversations — a numerical representation of meaning. For example, "Why does pressure increase when we go deeper into water?" asked earlier and "Why is pressure greater at the bottom of a liquid?" asked later use different words but express closely related concepts; semantic search can identify that relationship. This gives the AI a form of long-term academic memory.

Rolling Summary

Recent context can be compressed into a short summary so important information is retained without continuously passing large amounts of conversation history.

Consider a student studying mathematics over several weeks — asking about derivatives on Monday, rate of change on Thursday, and slope-as-derivative two weeks later. A system with relevant learning memory can connect these conversations, so the AI is no longer simply answering the latest question but working within a larger history of the student's learning.

Current Learning Focus

KnowLiq also maintains a current learning focus (for example: Bernoulli Equation, Continuity Equation, Hydrostatic Pressure) so the tutor keeps the session focused, preventing it from becoming an unrestricted conversation where the student jumps randomly between unrelated concepts.

24–29. Diagnostic Testing & Evaluation

When Does KnowLiq Test the Student?

Teaching alone is not enough — at some point the student needs to demonstrate understanding. After enough interaction within a learning session, KnowLiq can determine that the student is ready for a diagnostic test. Importantly, the test is connected to what the student actually studied, not a random collection of questions.

How KnowLiq Generates Tests

The test-generation pipeline uses the student's recent learning session, looking at concepts discussed, current learning topics, recent conversation, possible confusion, knowledge gaps, and topics allowed for the assessment. A lightweight model first summarizes the recent learning conversation, and then GPT-5.6 Luna generates the diagnostic test. The current system generates six questions, each containing question text, four answer options, the correct option, a reasoning prompt, and target topics.

Tests Adapt to the Subject

For technical subjects such as mathematics and engineering, when the learning conversation indicates numerical problem solving, the test can contain numerical multiple-choice questions with numerical answer options, and the student can be asked to provide reasoning about formulas or solution steps. For theory-oriented subjects, the system generates deeper conceptual multiple-choice questions — making the assessment more appropriate to the nature of the subject.

Why the Test Includes Reasoning

Consider two students who both choose the correct answer: Student A uses the correct formula and reaches the answer logically, while Student B guesses correctly. Both technically got the question right, but they do not demonstrate the same level of understanding. This is why KnowLiq includes reasoning prompts for suitable technical questions — the goal is to understand not only "Did you get the answer?" but also "Do you understand why the answer is correct?"

AI-Based Evaluation

After the student submits the test, GPT-5.6 Luna evaluates the student's performance at the subtopic level. The result includes the subtopic, remarks, and retention state, calibrated as:

Green

Correct and supported by sound reasoning.

Yellow

The main idea is understood, but reasoning or some concepts still need improvement.

Red

The answer is wrong, incomplete, or demonstrates flawed reasoning.

The Test Changes the Student's Learning State

The test result feeds back into the student's workspace. For example, Hydrostatic Pressure might move from *Untested* to **Red**, giving KnowLiq a reason to give this topic more attention. Another topic, such as Bernoulli Equation, might move from *Untested* to **Green**, indicating demonstrated understanding. This creates a feedback loop between assessment and future learning.

30. The Complete Learning Loop

1. Material – the student provides academic resources
↓
2. Knowledge Extraction – AI identifies topics, subtopics, importance, marks, frequency
↓
3. Workspace – KnowLiq creates a persistent academic workspace
↓
4. Prioritization – the system selects topics that deserve attention
↓
5. Personalized Tutoring – taught according to learning style, language, field, response preference
↓
6. Memory – relevant previous learning interactions can be retrieved
↓
7. Assessment – KnowLiq generates a diagnostic test based on the actual learning session
↓
8. Evaluation – AI evaluates answers and reasoning
↓
9. Retention – subtopics receive updated learning states
↓
10. Adaptation – updated states influence future topic priorities
↓
11. Repeat – the learning process continues

31. A Complete Example

Consider a student preparing for a Fluid Mechanics examination. They upload several past papers. KnowLiq analyzes them and creates a knowledge map identifying Fluid Properties, Hydrostatics, Continuity, Bernoulli Equation, and Momentum Equation — also noting which topics appear more frequently and carry more marks.

The student starts learning. KnowLiq prioritizes Bernoulli Equation and Hydrostatics. The student selects Visual & Analogy Mode with Detailed responses, so the tutor provides intuitive, detailed explanations. The student asks questions about Bernoulli's equation; the system keeps the current learning focus, and relevant previous conversations can be retrieved when needed.

After sufficient learning interaction, KnowLiq generates a six-question diagnostic test. The student submits answers, and the evaluation determines: Bernoulli Equation → **Green**, Hydrostatics → **Yellow**, Pressure Head → **Red**. The next learning session can therefore prioritize the weaker areas, and the process continues.

32. What Makes This Different From a Simple AI Chatbot?

A simple chatbot can answer "Explain thermodynamic entropy." KnowLiq goes further, considering: What subject is this? Is entropy part of the current learning focus? Has the student discussed entropy before? What related concepts have they already studied? What is their preferred learning style? What language do they prefer? Do they want a detailed or normal response? Have they demonstrated understanding? Should this topic be tested? What happened when they were tested? Should this topic receive more attention next time?

This is the difference between a question-answering interface and a learning-state system.

33–37. KnowLiq vs General-Purpose AI

KnowLiq should not be understood as a replacement for every general AI assistant. Systems such as ChatGPT, Gemini, and Claude are much broader — they can help with writing, coding, research, brainstorming, general knowledge, image understanding, analysis, education, productivity, and many other tasks. KnowLiq has a much narrower purpose: personalized academic learning. This specialization allows the product architecture to revolve around learning rather than general AI assistance.

KnowLiq vs ChatGPT

ChatGPT is a highly capable general-purpose AI system, and its current Study Mode can guide students through learning using questions, step-by-step explanations, understanding checks, uploaded materials, and practice activities. It would be inaccurate to say KnowLiq is simply "an AI tutor while ChatGPT is not" — the more meaningful comparison is workflow. KnowLiq is specifically organized around academic workspaces,

topic/subtopic mapping, learning-state tracking, semantic learning memory, retention states, session-based assessment, and feedback-driven prioritization, while ChatGPT has a much broader product scope.

KnowLiq vs Gemini

Gemini has developed strong education-focused capabilities — its Study Notebooks can work with learning materials, provide diagnostic quizzes, create personalized lessons, identify knowledge gaps, and support progress-oriented study, making it one of the strongest comparisons for KnowLiq. KnowLiq's strongest distinction is its specific architecture around Subtopics → Retention → Assessment → Future Priority, explicitly maintaining a learning state for the student's academic workspace, while Gemini benefits from Google's enormous ecosystem and broad AI capabilities.

KnowLiq vs Claude

Claude also has education-focused capabilities, including Learning Mode, which emphasizes guided reasoning and critical thinking rather than simply giving students the final answer — overlapping with KnowLiq's Interactive Socratic Mode. The distinction is not that KnowLiq is the only system capable of Socratic learning, but that KnowLiq combines that tutoring approach with persistent academic workspaces, topic prioritization, learning memory, explicit retention states, diagnostic testing, AI evaluation, and feedback-driven study planning.

Competitive Comparison

Capability	KnowLiq	ChatGPT	Gemini	Claude
General AI assistance	Focused on education	Very broad	Very broad	Very broad
AI tutoring	Yes	Yes	Yes	Yes
Guided learning	Yes	Yes	Yes	Yes
Uploaded academic material	Yes	Yes	Yes	Yes
Academic knowledge mapping	Core workflow	Not primary focus	Study-focused	Not primary focus
Persistent subject workspace	Core workflow	Broader workspace/ memory	Study-oriented	Education-oriented
Semantic learning memory	Core workflow	Memory/context capabilities	Memory/context capabilities	Context capabilities
Explicit subtopic retention states	Core workflow	Not central workflow	Progress/ adaptation features	Not central workflow
Diagnostic assessment	Yes	Yes	Yes	Yes

Assessment based on recent learning	Core workflow	Possible via broader features	Adaptive study capabilities	Guided learning
Test results influence topic priority	Core workflow	Not central workflow	Adaptive learning features	Not central workflow
Learner-type selection	Yes	Guided learning	Adaptive study	Learning Mode
Multiple response structures	Yes	Yes	Yes	Yes
Multilingual learning	Yes	Yes	Yes	Yes
Task-oriented model strategy	Stella / Astra / Auto	Multiple models	Multiple models	Multiple models

The goal of this comparison is not to claim that KnowLiq is universally better — the major AI platforms have enormous advantages in scale, research, model capabilities, tools, and general-purpose functionality. KnowLiq's strongest position is different: it is a focused academic system built around the student's persistent learning state.

38–41. Where KnowLiq Helps & Personalization

Where KnowLiq Can Be Especially Helpful

- **Exam Preparation** — turns academic resources and past papers into a structured learning environment
- **Finding Weak Areas** — diagnostic assessments provide evidence about understanding rather than relying entirely on the student's judgment
- **Long-Term Learning** — persistent workspaces and learning memory become increasingly useful over weeks or months
- **Technical Education** — supports mathematics, engineering, physics, chemistry, and numerical problem solving
- **Revision** — retention information helps identify concepts needing additional attention
- **Personalized Learning** — students choose language, learner type, response structure, field of study, and AI model

Personalization in KnowLiq

Not every student should receive exactly the same AI experience. Two students may study the same subject but prefer completely different approaches — for example, Student A may want Urdu + Visual & Analogy + Detailed, Student B may want English + Socratic + Normal, and Student C may want Hindi + To-The-Point + Normal. The academic material may be identical, but the tutoring experience can differ.

Personalization Has Multiple Dimensions

- **What to learn** — determined through topics, importance, learning state, and prioritization

- **How to learn** — determined through learner type (Visual & Analogy, Interactive Socratic, To-The-Point)
- **What language to use** — selected from supported language options
- **How much detail to provide** — Detailed or Normal
- **Which AI model to use** — Stella, Astra, or Auto

Cost Efficiency Is Also Part of the Design

AI systems can become expensive when every interaction uses the most powerful model and generates unnecessarily long responses. KnowLiq includes resource-conscious decisions: Astra uses GPT-4o-mini for lighter interactions, Auto currently defaults to GPT-4o-mini, Normal response mode is designed to reduce cost by approximately 30%, and more capable models are reserved for situations where stronger reasoning is useful. This pursues a balance between learning quality, response efficiency, and operational cost.

42–43. The Technology Behind KnowLiq

Frontend

Built using React, Vite, React Router, Tailwind CSS, Axios, Firebase, React Markdown, KaTeX, and Lucide React — providing the student's interactive learning environment.

Backend

Built using Node.js, Express, MongoDB, Mongoose, Firebase Admin, OpenAI APIs, Multer, and PDF processing — coordinating the learning workflow and persistent data.

Database and Persistent Data

KnowLiq stores important learning information in MongoDB, with data models including Users, Workspaces, Chats, Tests, and Past Papers. The workspace stores academic structure, chat records store learning interactions, tests store assessment information, and evaluation results update the learning state — together forming the persistent learning system.

44–45. Behind a Chat Message and a Test

What Happens Behind One Chat Message?

When the student sends a question, the process is more than "Question → AI → Answer." Conceptually:

```
Student Question
  ↓
Identify selected AI model
  ↓
Load student and workspace
  ↓
```

```
graph TD; A[Check current learning state] --> B[Determine current learning focus]; B --> C[Retrieve relevant learning memory]; C --> D[Build personalized tutoring context]; D --> E[Send the request to the selected AI model]; E --> F[Receive structured tutor response]; F --> G[Update learning state]; G --> H[Save the conversation]; H --> I[Return the answer to the student];
```

What Happens Behind a Test?

Similarly, a KnowLiq test is not simply "Click 'Generate Quiz' → Random Questions." Instead:

```
graph TD; A[Current Learning Session] --> B[Retrieve relevant conversations]; B --> C[Understand concepts discussed]; C --> D[Identify possible gaps]; D --> E[Generate targeted questions]; E --> F[Restrict questions to relevant learning topics]; F --> G[Student answers]; G --> H[AI evaluates answers and reasoning]; H --> I[Assign retention states]; I --> J[Update workspace];
```

↓
Use updated states for future learning

46–50. The Closed Loop, Vision & Final Perspective

Why the Closed Loop Matters

A system that teaches but never tests cannot reliably determine whether the student understood. A system that tests but does not remember the result means the next session starts almost from zero. A system that remembers results but does not use them to change study priorities has memory of limited value. KnowLiq connects all three — Teaching → Assessment → Adaptation — creating the closed learning loop.

KnowLiq's Strongest Concept

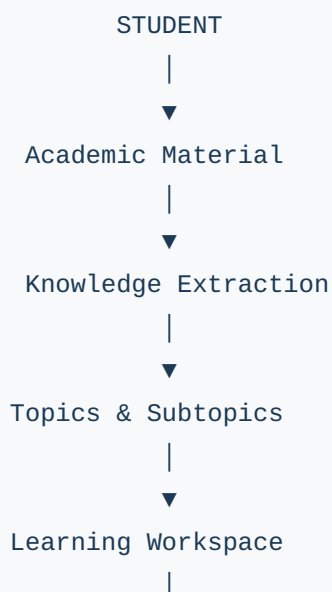
Learning state drives future learning. The system does not simply collect information about the student — it attempts to use that information to make the next learning interaction more relevant.

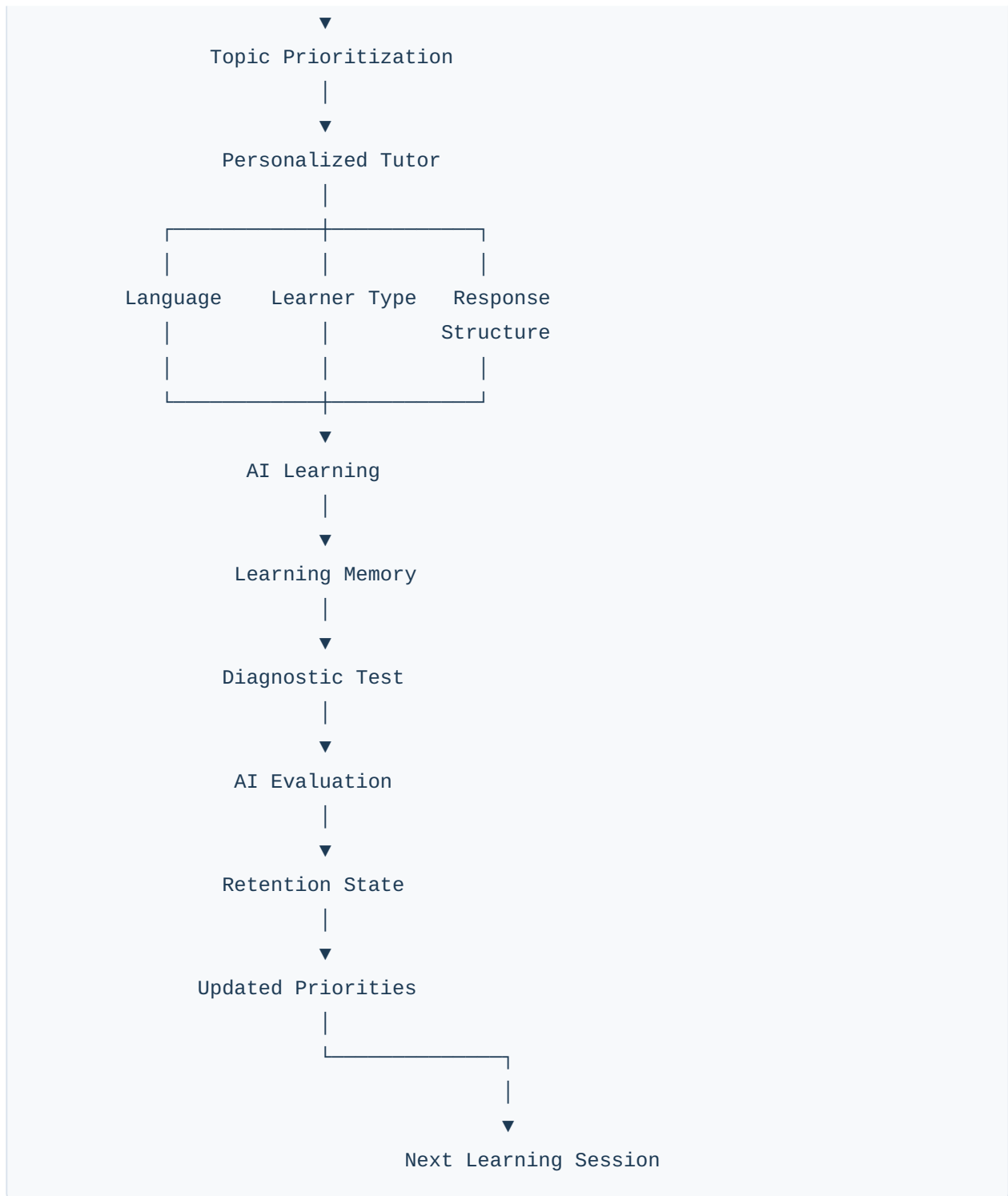
Red → Needs attention. Yellow → Needs reinforcement. Green → Strong / review when due. That information becomes part of the next prioritization decision.

What KnowLiq Is Ultimately Trying to Build

The long-term vision is not simply another chatbot — it is a system that can continuously maintain a model of the student's academic journey: what the student needs to learn, what they have studied, what they understand, where they are struggling, what they have forgotten, what should be revised, what should be tested, and what they should learn next — using this information to continuously adapt the learning experience.

The Vision in One Pipeline





Final Perspective

AI has already made it possible for students to ask almost any question and receive an explanation. The next challenge is making AI understand the learning process around those questions. KnowLiq takes academic material and turns it into a structured learning environment: it identifies topics and subtopics, considers academic importance, prioritizes what the student should learn, and provides an AI tutor that adapts to the student's preferred learner type, language, field of study, response structure, and selected model. It remembers relevant learning history, generates tests based on what the student actually studied, evaluates not only

answers but understanding and reasoning, assigns retention states, and uses those retention states to influence what the student should focus on next.

Learn → Remember → Test → Evaluate → Adapt → Learn Again

KnowLiq is therefore not intended to be simply another place where students ask an AI questions. It is designed as a persistent, personalized academic learning system where the AI can participate in the student's learning journey rather than only responding to individual prompts.

The ultimate goal is straightforward: help students spend less time figuring out how to study and more time actually learning.

Appendix. Interface & Display Settings — Dark Theme Toggling

INTERFACE PERSONALIZATION

Alongside academic personalization (language, learner type, response structure, and AI model), KnowLiq's interface itself supports a **dark theme toggle**, giving students control over the visual appearance of the learning workspace in addition to how the content is taught.

Dark Theme

A darker interface palette intended for reduced eye strain during extended study sessions, low-light environments, or personal visual preference. Students can switch to dark mode for long revision sessions at night or during exam preparation marathons.

Light Theme

The default, brighter interface palette, better suited to well-lit environments and daytime study sessions.

The theme toggle is a display-only preference — it changes the visual appearance of the workspace, chat, and dashboard, and does not affect the underlying learning logic, topic prioritization, retention states, or AI tutoring behavior. It sits alongside the platform's other personalization controls (language, learner type, response structure, and model selection) as part of giving each student a study environment suited to their own preferences.